

3(a)(i)	one mark for units one mark per column <table border="1"> <tr> <td></td><td>$V_{\max}$ $\mu\text{mol min}^{-1}$</td><td>K_m mmol dm^{-3} ;</td></tr> <tr> <td>J</td><td>60</td><td>0.2</td></tr> <tr> <td>K</td><td>50 ;</td><td>0.2 ;</td></tr> </table>		$V_{\max}$ $\mu\text{mol min}^{-1}$	K_m mmol dm^{-3} ;	J	60	0.2	K	50 ;	0.2 ;	3
	$V_{\max}$ $\mu\text{mol min}^{-1}$	K_m mmol dm^{-3} ;									
J	60	0.2									
K	50 ;	0.2 ;									
3(a)(ii)	<i>K compared with J</i> max 3 from mp4–mp8 any four from: 1. rate of reaction is lower at all substrate concentrations ; 2. <i>ref. to</i> greater difference as substrate concentration increases ; 3. rate of reactions for J and K compared at two substrate concentrations to support mp1 or mp2 or manipulated data at two substrate concentrations to support mp2 ; 4. less efficient / acts more slowly / AW ; 5. active site is still binding substrate ; 6. <i>idea that</i> active site is less effective at catalysing reaction 7. <i>or</i> enzyme has lower affinity for substrate ; 8. (replaced / changed) amino acid is important in, shape / structure, of active site ; 9. AVP ; e.g. how the mutation may have changed interaction between substrate and, active / catalytic, site e.g. <i>ref. to</i> hydrogen bonding / transfer of electrons	4									
3(b)	each polypeptide shows primary, secondary and tertiary structure ; A description of levels of structure both (forms of G6PD / enzyme) composed of more than one polypeptide (so have quaternary structure) ;	2									